Roots of polynomial functions

1 Determine whether the following values are zeros of the function $f(x) = x^4 - 7x^3 + 12x^2 + 4x - 16$:

a $x = 5$ **b** $x = 4$ **c** $x = -1$ **d** $x = 1$

2 For each of the following functions:

i State the zeros of the function. **ii** State the multiplicity of each zero.

a $f(x) = (x+6)^3 (x-3)(x+9)$

b $f(x) = (x+1)^2 (x-6)(x+2)^5$

c $f(x) = -3(x-6)^4 (x+2)^5 x^3$

d $f(x) = x^3 - 3x^2 - 5x + 15$

3 Determine whether each of the following equations have a root of multiplicity 4:

a $(x+8)^4 (x+3) = 0$

b $(x+8)(x+3)^3 = 0$

c $(x+8)(x+3) = 0$

d $(x+8)^3 (x+3) = 0$

4 Determine whether each of the following equations have a root of multiplicity 3:

a $(x+1)\left(x^2+2x+1\right)(x+3) = 0$

b $(x+1)^3 (x+3) = 0$

c $(x+1)^2 (x+1)(x+3) = 0$

d $(x+1)(x-9)(x+3) = 0$

e $(x+2)(x-8)^3 (x-1) = 0$

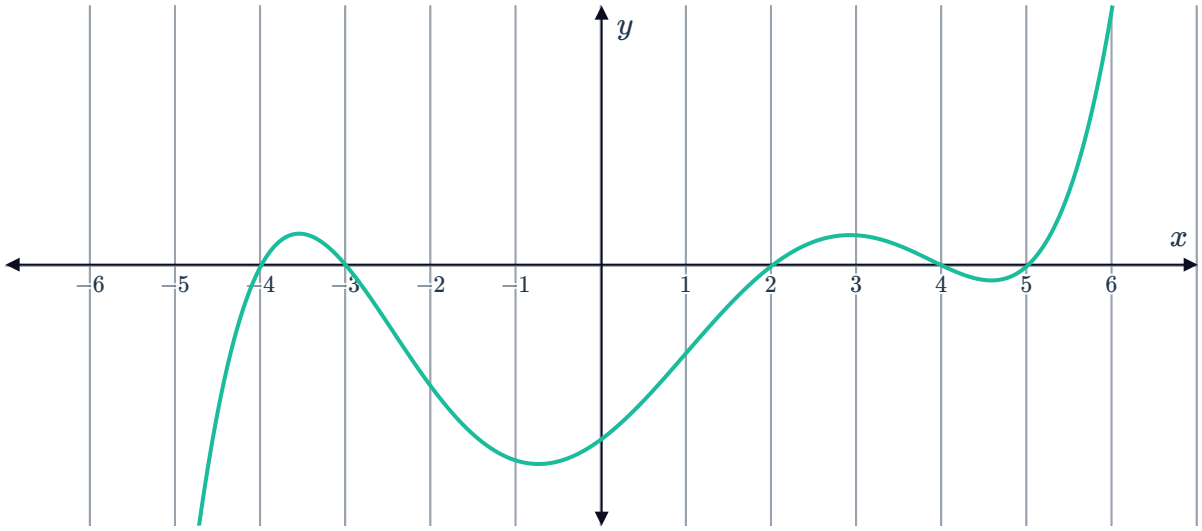
f $\left(x^3+1\right)(x-2)(x+1) = 0$

g $(x+3)(x-1)^2 (x-3) = 0$

h $\left(x^2-2x+1\right)(x-1)(x+9)^3 = 0$

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5 Consider the following graph of a function $f(x)$:



a State all the linear factors of $f(x)$.

b Write an expression for a polynomial $h(x)$ with a leading coefficient of 1, as a product of linear factors that has exactly the same roots as $f(x)$.

c Determine whether the following has roots which include all the roots of $f(x)$:

i $2(x+4)(x-5)(x+3)(x-4)(x-2)$

ii $(x+4)(x+3)(x-2)^2(x-5)$

iii $4(x-4)(x-3)(x+2)(x+4)(x+5)$

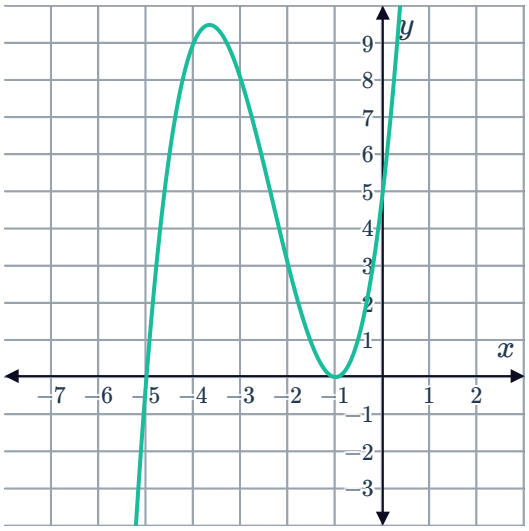
iv $(x+4)(x+3)(x-2)(x-4)(x-5)(x+7)(x-11)$

d Apart from $f(x)$, how many other polynomials are there with the same roots and the same graph as $f(x)$?

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Equations of polynomial functions

6 Consider the following graph of a cubic function:

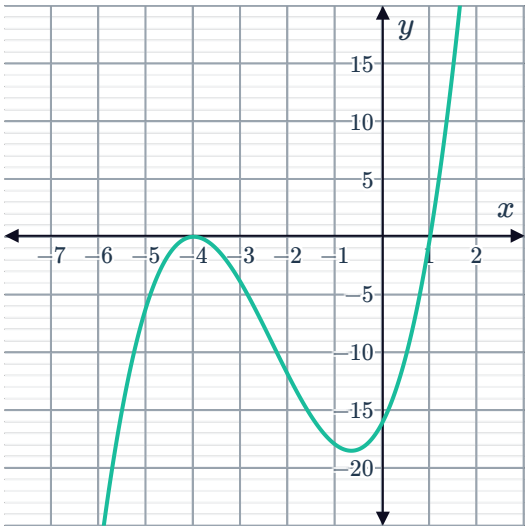


a State the x-intercepts of the function.

b State the x-intercept that has a double root.

c State the equation of the cubic function.

7 Find the equation of the graphed cubic function:



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8 Find the equation of the following cubic functions in standard form:

a

- Has real coefficients
- Zeros at $x = 5$, $x = -3$, $x = 1$
- When $x = -2$, $y = -84$.

b

- Has real coefficients
- Zeros at $x = 0$, $x = 2$, $x = 5$
- When $x = -1$, $y = 72$.

c

- Has real coefficients
- Zeros at $x = 0$, $x = 6$, $x = -6$
- When $x = -4$, $y = 20$.

d

- Has real coefficients
- Zeros at $x = -5$, $x = 4$, $x = -1$
- When $x = 2$, $y = -126$.

e

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = 5$
- When $x = 2$, $y = 18$.

f

- Has real coefficients
- Zeros at $x = -1$, $x = -4$, $x = 3$
- When $x = -5$, $y = 64$.

g

- Has real coefficients
- Zeros at $x = -3$, $x = -1$, $x = -4$
- When $x = -2$, $y = -10$.

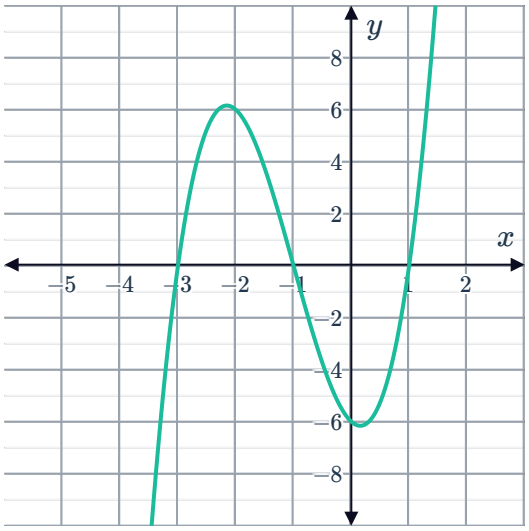
h

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = -2$
- When $x = 3$, $y = 50$.

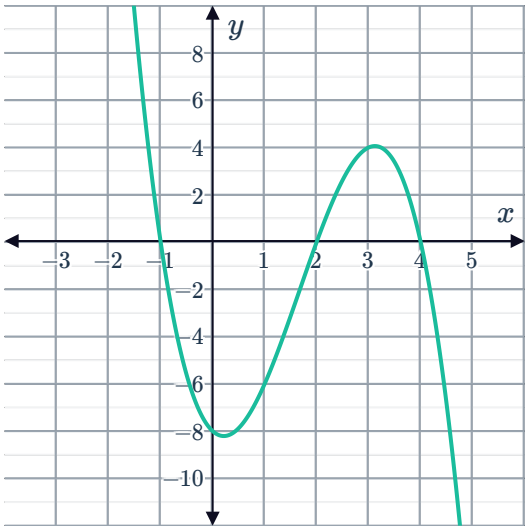
9 Evan graphed a cubic function, $f(x) = ax^3 + bx^2 + cx + d$, and determined the roots of $f(x)$ to be ± 4 and 5. Find the value of b if $a = 1$.

10 Find the equation of the following graphed functions in factored form:

a

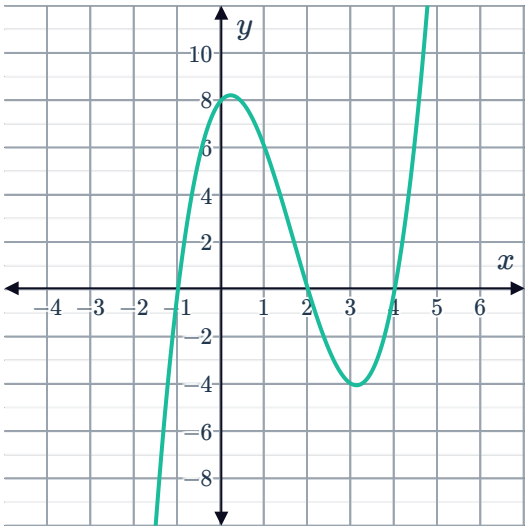


b

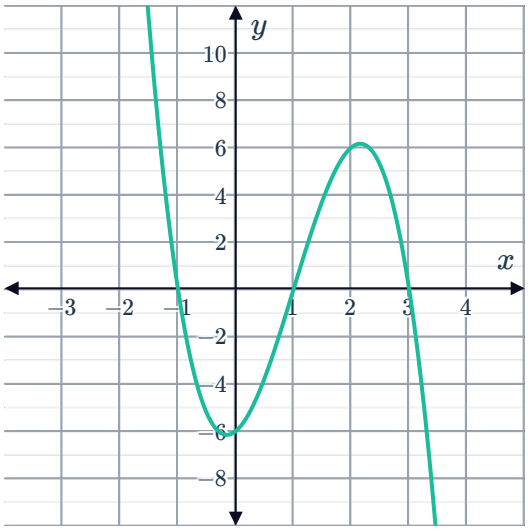


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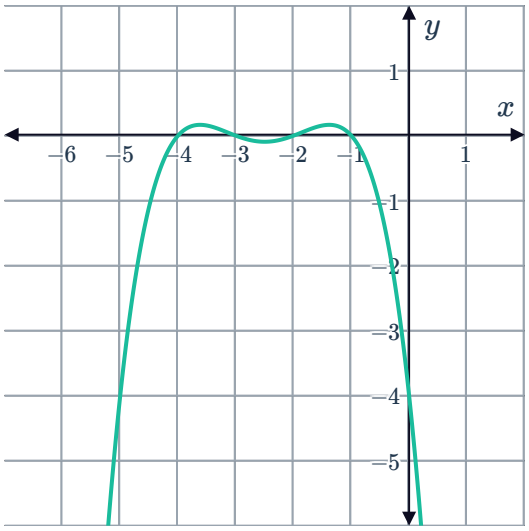
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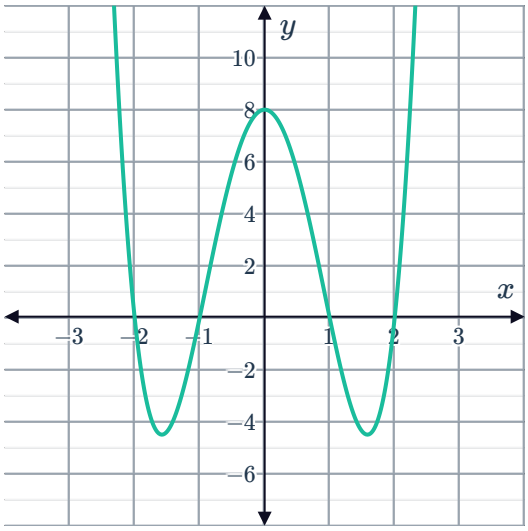
d



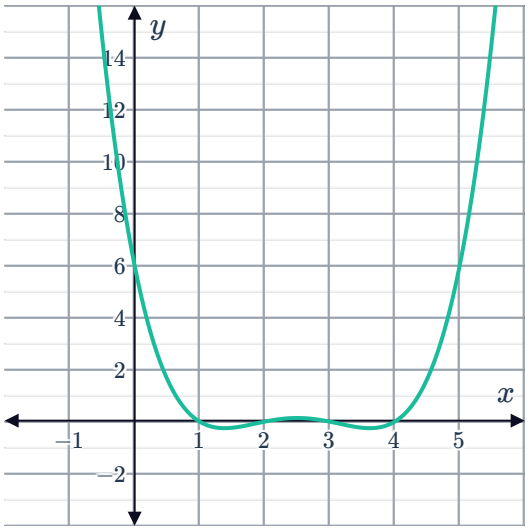
e



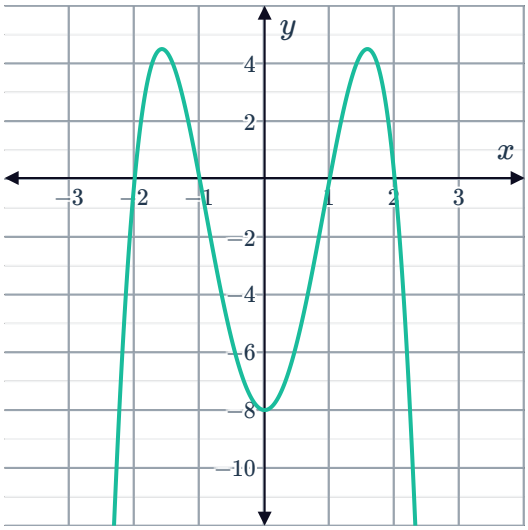
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